Reg. No. | | | | | | | | | | | [17EC42]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

ELECTRO MAGNETIC FIELD THEORY

Time: 3 Hours **Maximum Marks:70**

Instructions:

i. **Answer any FIVE full questions by choosing at least TWO questions from each part.**

ii. **Any missing data may be suitably assumed.**

PART - A

1. a. Define 'Electric field intensity' due to a point charge. Plot the variation of E with respect to distance along radial direction due to a point charge. **06**

   b. Evaluate both the sides of the divergence theorem for the field $\vec{D} = \frac{2}{z^2}(yz\vec{a}_x + xz\vec{a}_y - 2xy\vec{a}_z)$ C/m^2 and the volume enclosed by $2 < x, y, z < 3$ in free space. **08**

2. a. Derive and discuss $\vec{E} = -\nabla V$. **06**

   b. Given the potential field, $V = 2x^2y - 5z$, find the potential *V*, the electric field intensity $\vec{E}$, the direction of $\vec{E}$, the electric flux density $\vec{D}$ and the volume charge density ρ_V at P(-4,3,6) in free space. **08**

3. a. State and explain Biot-Savart's law. Using this, find the magnetic flux density at the centre of a circular current loop of radius '*a*' m. **08**

   b. Verify the Stoke's theorem for the field $\vec{H} = 6xy\vec{a}_x - 3y^2\vec{a}_y$ A/m and the rectangular path around the region, $2 \le x \le 5$, $-1 \le y \le 1$, $z = 0$. Let the positive direction of $\vec{ds}$ be $\vec{a}_z$. **06**

4. a. With usual notations, derive the equation for magnetic force between two differential current elements. **06**

   b. Find the torque vector on a square loop having corners (-2, -2, 0), (2, -2, 0), (2, 2, 0) and (-2, 2, 0). **04**

      i. About the origin by $\vec{B} = 0.4\vec{a}_x$T

      ii. About the origin by $\vec{B} = 0.6\vec{a}_x - 0.4\vec{a}_y$T

   c. Derive Lorentz force equation. **04**

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PART - B

5. a. State and explain Faraday's laws and hence show that $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$. **05**

b. List Maxwell's equations in point and integral forms for- **06**

i. Static fields

ii. Time varying fields

c. If $\vec{D} = 10x\vec{a}_x - 4y\vec{a}_y + kz\vec{a}_z\ \mu C/m^2$ and $\vec{B} = 2\vec{a}_y\ mT$, find the value of k to satisfy the Maxwell's equations for region $\sigma = 0$, $\rho_v = 0$. **03**

6. a. Obtain the solution of wave equation for uniform plane wave propagating in free space. **08**

b. An $\vec{E}$ field in free space is given as $\vec{E} = 800\cos(10^8 t - \beta y)\,\vec{a}_z$ V/m. Find **06**

i. β

ii. λ

iii. $\vec{H}$ at $p(0.1, 1.5, 0.4)$ at time t = 8 nsec.

7. a. Discuss reflection of uniform plane waves at the boundary between two media for normal incidence. Derive expressions for reflection co-efficient and transmission co-efficient. **10**

b. Determine reflection co-efficient and transmission co-efficient values at the interface between medium1 and medium2 having wave impedances 100 Ω and 300 Ω respectively. **04**

8. a. Explain various frequency bands and their application areas in electromagnetic spectrum. **10**

b. List hazardous effects of electromagnetic waves. **04**